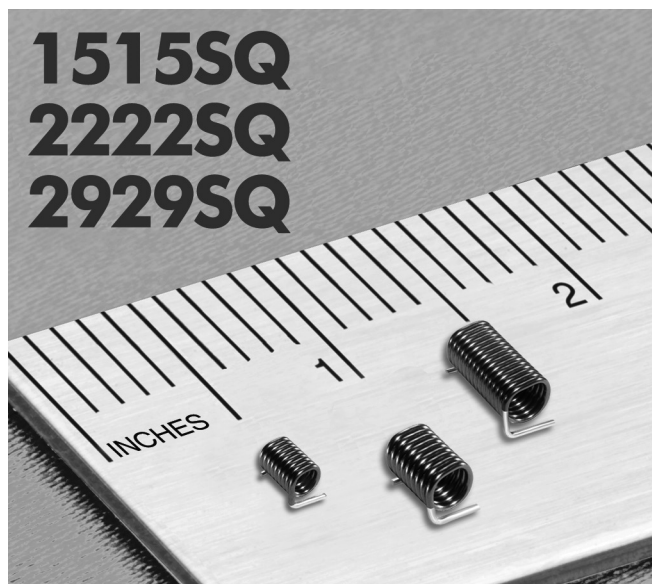




Square Air Core Inductors

1515SQ
2222SQ
2929SQ



- Excellent Q factors – up to 230 at 400 MHz!
- Current handling as high as 5.7 Amps
- Inductance values from 47 to 500 nH
- Flat top and bottom for reliable pick and place and mechanical stability

Designer's Kit C438 contains 10 each of all 5% parts

Terminations RoHS compliant tin-silver over copper

Ambient temperature -40°C to +125°C with Irms current

Maximum part temperature +145°C (ambient + temp rise).

Storage temperature Component: -40°C to +145°C.
Tape and reel packaging: -40°C to +80°C

Resistance to soldering heat Max three 40 second reflows at +260°C, parts cooled to room temperature between cycles

Temperature Coefficient of Inductance (TCL) +5 to +70 ppm/°C

Moisture Sensitivity Level (MSL) 1 (unlimited floor life at <30°C / 85% relative humidity)

PCB washing Tested to MIL-STD-202 Method 215 plus an additional aqueous wash. See [Doc787_PCB_Washing.pdf](#).

Part number ¹	Inductance ² (nH)	Percent tolerance ³	Q ⁴ typ	Test frequency (MHz)	SRF min ⁵ (GHz)	DCR max (mOhm)	Irms ⁶ (A)
1515SQ-47N_E	47	5,2	230	400	1.87	6.35	4.9
1515SQ-68N_E	68	5,2	230	400	2.13	8.60	5.5
1515SQ-82N_E	82	5,2	230	400	1.79	9.40	5.6
2222SQ-90N_E	90	5,2	140	50	1.15	5.50	5.0
2222SQ-111_E	110	5,2	140	50	1.00	6.50	5.7
2222SQ-131_E	130	5,2	140	50	1.00	7.50	5.4
2222SQ-161_E	160	5,2	140	50	1.00	8.25	5.7
2222SQ-181_E	180	5,2	140	50	1.10	9.50	5.0
2222SQ-221_E	220	5,2	140	50	1.00	11.0	5.0
2222SQ-271_E	270	5,2	140	50	0.800	12.5	4.3
2222SQ-301_E	300	5,2	150	50	0.720	13.8	3.7
2929SQ-331_E	330	5,2	180	50	0.660	12.5	4.7
2929SQ-361_E	360	5,2	180	50	0.620	13.5	4.5
2929SQ-391_E	390	5,2	180	50	0.590	14.5	4.4
2929SQ-431_E	430	5,2	180	50	0.550	15.5	4.2
2929SQ-501_E	500	5,2	180	50	0.500	16.5	4.3

1. When ordering, specify **tolerance**, **termination** and **packaging** codes:

1515SQ-82NJE**C**

Tolerance: G = 2% J = 5%

(Table shows stock tolerances in bold.)

Termination: E = RoHS compliant tin-silver (96.5/3.5) over copper.

Special order, added cost:

T = RoHS tin-silver-copper (95.5/4/0.5) over copper or S = non-RoHS tin-lead (63/37) over copper.

Packaging: C = 7" machine-ready reel. EIA-481 embossed plastic tape. Quantities less than full reel available: in tape (not machine ready) or with leader and trailer (\$25 charge).

D = 13" machine-ready reel. EIA-481 embossed plastic tape. Factory order only, not stocked.

B = Less than full reel. In an effort to simplify our part numbering system, Coilcraft is eliminating the need for multiple packaging codes. When ordering, simply change the last letter of your part number from B to C.

2. Inductance measured at specified test frequency, 0.1 Vrms, 0 A using an Agilent/HP 4287A LCR meter with the following Coilcraft test fixtures: CCF1191C for 1515SQ; CCF1197A for 2222SQ and 2929SQ.

3. Tolerances in bold are stocked for immediate shipment.

4. Q measured at specified test frequency, using an Agilent/HP 4291A impedance analyzer.

5. SRF measured using an Agilent/HP 8753 network analyzer and the following Coilcraft test fixtures: CCF1143 for 1515SQ; CCF1199 for 2222SQ-90N and 2222SQ-161; CCF1200 for 2222SQ-301 and 2929SQ-501.

6. Current that causes a 20°C temperature rise from 25°C ambient. This information is for reference only and does not represent absolute maximum ratings.

7. Electrical specifications at 25°C.

Refer to Doc 362 "Soldering Surface Mount Components" before soldering.



www.coilcraft.com

US +1-847-639-6400 sales@coilcraft.com

UK +44-1236-730595 sales@coilcraft-europe.com

Taiwan +886-2-2264 3646 sales@coilcraft.com.tw

China +86-21-6218 8074 sales@coilcraft.com.cn

Singapore + 65-6484 8412 sales@coilcraft.com.sg

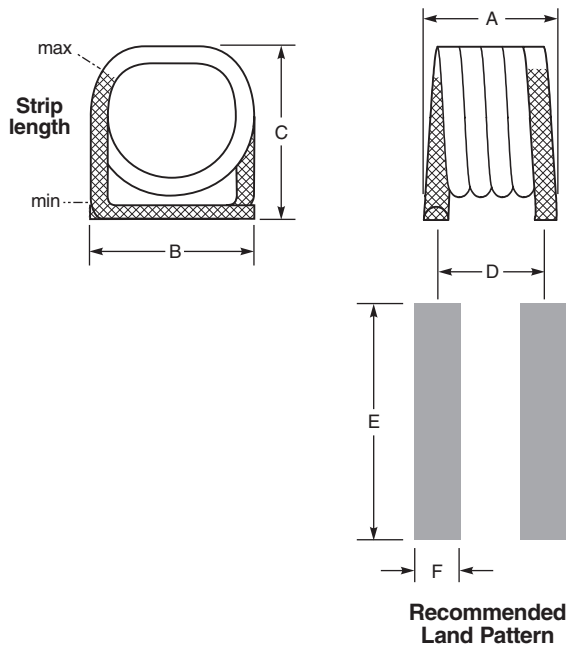
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Square Air Core Inductors – 1515SQ, 2222SQ, 2929SQ



Packaging:

1515SQ-47N 500/7" reel; 2000/13" reel Plastic tape: 12 mm wide, 0.26 mm thick, 8 mm pocket spacing, 4.06 mm pocket depth

1515SQ-68N 500/7" reel; 2000/13" reel Plastic tape: 12 mm wide, 0.26 mm thick, 8 mm pocket spacing, 4.06 mm pocket depth

1515SQ-82N 500/7" reel; 2000/13" reel Plastic tape: 12 mm wide, 0.26 mm thick, 8 mm pocket spacing, 4.06 mm pocket depth

2222SQ-90N 250/7" reel; 750/13" reel Plastic tape: 16 mm wide, 0.32 mm thick, 12 mm pocket spacing, 5.97 mm pocket depth

2222SQ-111 250/7" reel; 750/13" reel Plastic tape: 16 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.10 mm pocket depth

2222SQ-131 250/7" reel; 750/13" reel Plastic tape: 16 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.10 mm pocket depth

2222SQ-161 250/7" reel; 750/13" reel Plastic tape: 16 mm wide, 0.33 mm thick, 12 mm pocket spacing, 5.97 mm pocket depth

2222SQ-181 250/7" reel; 750/13" reel Plastic tape: 16 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.10 mm pocket depth

2222SQ-221 250/7" reel; 750/13" reel Plastic tape: 24 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.10 mm pocket depth

2222SQ-271 250/7" reel; 750/13" reel Plastic tape: 24 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.10 mm pocket depth

2222SQ-301 250/7" reel; 750/13" reel Plastic tape: 24 mm wide, 0.33 mm thick, 12 mm pocket spacing, 5.97 mm pocket depth

2929SQ-331 150/7" reel; 600/13" reel Plastic tape: 24 mm wide, 0.40 mm thick, 12 mm pocket spacing, 6.99 mm pocket depth

2929SQ-361 150/7" reel; 600/13" reel Plastic tape: 24 mm wide, 0.40 mm thick, 12 mm pocket spacing, 7.75 mm pocket depth

2929SQ-391 150/7" reel; 600/13" reel Plastic tape: 24 mm wide, 0.45 mm thick, 12 mm pocket spacing, 7.75 mm pocket depth

2929SQ-431 150/7" reel; 600/13" reel Plastic tape: 24 mm wide, 0.45 mm thick, 12 mm pocket spacing, 7.75 mm pocket depth

2929SQ-501 150/7" reel; 600/13" reel Plastic tape: 24 mm wide, 0.33 mm thick, 12 mm pocket spacing, 7.49 mm pocket depth

Recommended pick and place nozzles

1515SQ: OD: 0.170 inch; ID: 0.059 inch

2222SQ, 47 – 200 nH: 3.5 mm × 7 mm (rubber). This nozzle is available [here](#).

2222SQ, 270 – 300 nH: 4.0 mm × 12 mm (rubber). This nozzle is available [here](#).

2929SQ: 4.0 mm × 12 mm (rubber). This nozzle is available [here](#).

Part number	A	B	C	D	E	F	Weight (mg)
1515SQ-47N	0.160±0.010 4.06 ±0.254	0.140 ±0.007 3.56 ±0.178	0.147 ±0.007 3.73 ±0.178	0.140 3.56	0.175 4.45	0.070 1.78	110
1515SQ-68N	0.210 ±0.010 5.33 ±0.254	0.140 ±0.007 3.56 ±0.178	0.147 ±0.007 3.73 ±0.178	0.190 4.83	0.175 4.45	0.070 1.78	145
1515SQ-82N	0.230 ±0.010 5.84 ±0.254	0.140 ±0.007 3.56 ±0.178	0.147 ±0.007 3.73 ±0.178	0.210 5.33	0.175 4.45	0.070 1.78	165
2222SQ-90N	0.205 ±0.015 5.21 ±0.381	0.215 ±0.010 5.46 ±0.254	0.224 ±0.010 5.69 ±0.254	0.175 4.45	0.255 6.48	0.085 2.16	280
2222SQ-111	0.250 ±0.015 6.35 ±0.381	0.220±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.230 5.84	0.265 6.73	0.085 2.16	330
2222SQ-131	0.265 ±0.015 6.73 ±0.381	0.220 ±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.245 6.22	0.265 6.73	0.085 2.16	371
2222SQ-161	0.290 ±0.015 7.37 ±0.381	0.220 ±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.260 6.60	0.265 6.73	0.085 2.16	425
2222SQ-181	0.320 ±0.015 8.13 ±0.381	0.220 ±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.290 7.37	0.265 6.73	0.085 2.16	460
2222SQ-221	0.390 ±0.015 9.91 ±0.381	0.220 ±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.360 9.14	0.265 6.73	0.085 2.16	550
2222SQ-271	0.460 ±0.015 11.68 ±0.381	0.220 ±0.010 5.59 ±0.254	0.224 ±0.010 5.69 ±0.254	0.420 10.67	0.265 6.73	0.085 2.16	640
2222SQ-301	0.470 ±0.015 11.94 ±0.381	0.225 ±0.010 5.72 ±0.254	0.224 ±0.010 5.69 ±0.254	0.440 11.18	0.265 6.73	0.085 2.16	650
2929SQ-331	0.405 ±0.015 10.29 ±0.381	0.295 ±0.010 7.49 ±0.254	0.285 ±0.010 7.24 ±0.254	0.375 9.53	0.325 8.26	0.09 2.29	855
2929SQ-361	0.445 ±0.015 11.30 ±0.381	0.295 ±0.010 7.49 ±0.254	0.285 ±0.010 7.24 ±0.254	0.415 10.541	0.325 8.26	0.09 2.29	928
2929SQ-391	0.485 ±0.015 12.32 ±0.381	0.295 ±0.010 7.49 ±0.254	0.285 ±0.010 7.24 ±0.254	0.455 11.56	0.325 8.26	0.09 2.29	990
2929SQ-431	0.520 ±0.015 13.21 ±0.381	0.295 ±0.010 7.49 ±0.254	0.285 ±0.010 7.24 ±0.254	0.490 12.45	0.325 8.26	0.09 2.29	1072
2929SQ-501	0.550 ±0.015 14.00 ±0.381	0.295 ±0.010 7.49 ±0.254	0.285 ±0.010 7.24 ±0.254	0.520 13.21	0.325 8.26	0.09 2.29	1160

All dimensions are in $\frac{\text{inches}}{\text{mm}}$.

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www.coilcraft.com

US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

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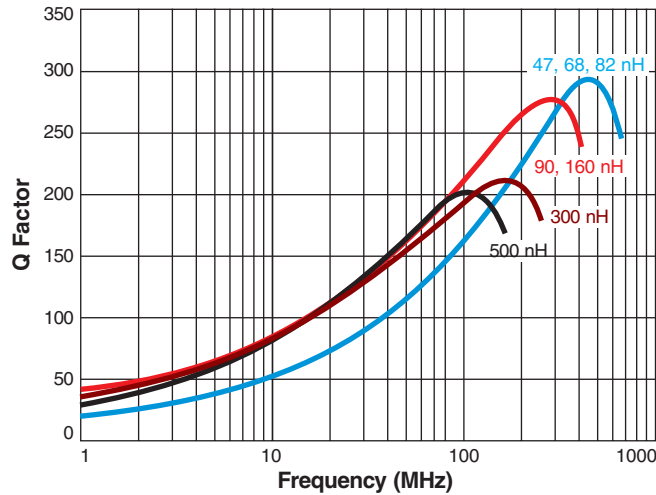
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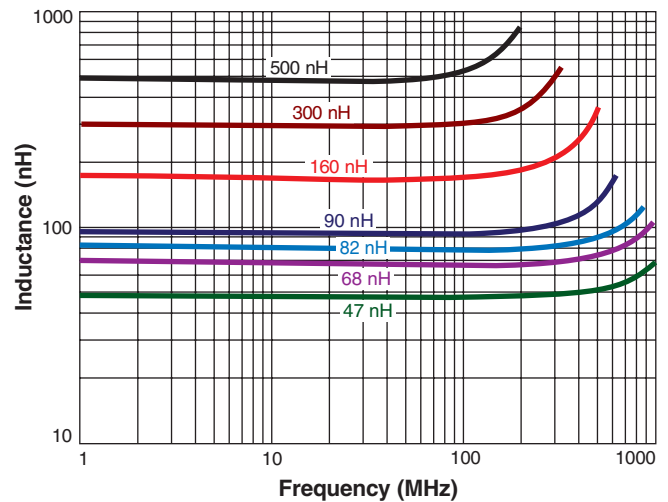


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Typical Q vs Frequency



Typical L vs Frequency



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US +1-847-639-6400 sales@coilcraft.com
UK +44-1236-730595 sales@coilcraft-europe.com
Taiwan +886-2-2264 3646 sales@coilcraft.com.tw
China +86-21-6218 8074 sales@coilcraft.com.cn
Singapore + 65-6484 8412 sales@coilcraft.com.sg

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